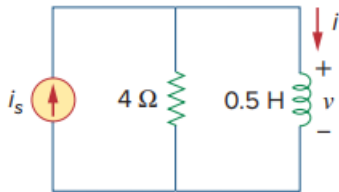


Problem

In the circuit in Fig. 7.126, i_s changes from 5 A to 10 A at $t = 0$; that is, $i_s = 5u(-t) + 10u(t)$. Find v and i .

Fig. 7.126:



Step-by-step solution

Step 1 of 7

Refer to figure 7.126 in the textbook.

Redraw the circuit for $t < 0$ as shown in figure 1.

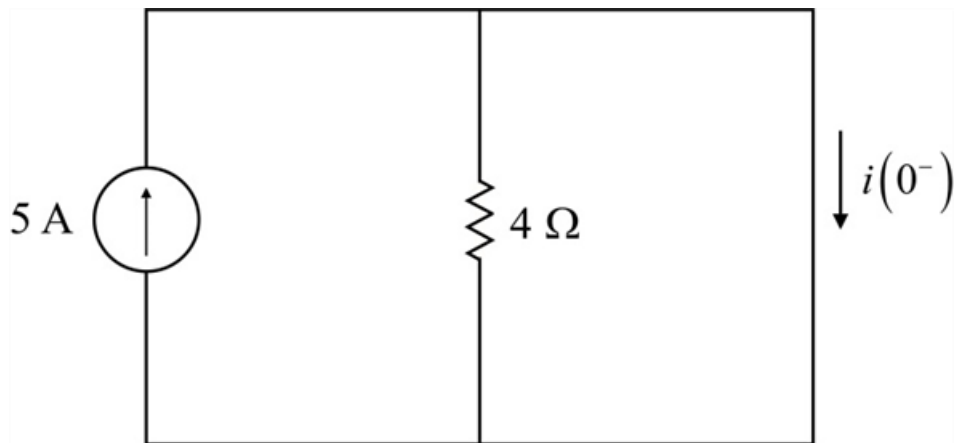


Figure 1

Step 2 of 7

Consider the circuit shown in figure 1.

$$i(0^-) = 5 \text{ A}$$

The inductor doesn't allow sudden changes in current.

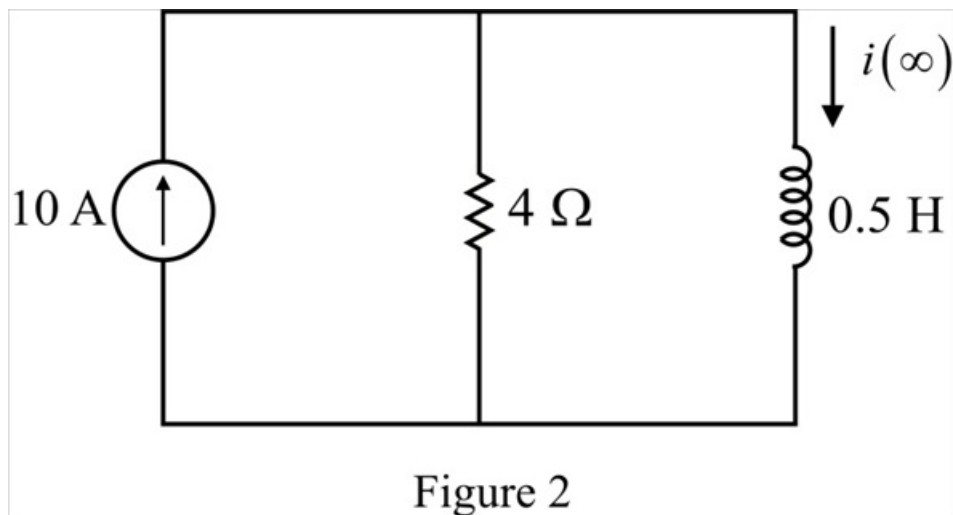
$$i(0^-) = i(0^+) = 5 \text{ A}$$

Therefore initial value of current i is,

$$i(0) = 5 \text{ A}$$

Step 3 of 7

Redraw the circuit for $t > 0$ as shown in figure 2.



Step 4 of 7

Calculate the steady state value of inductor current.

Apply the source transformation technique.

$$\begin{aligned} i(\infty) &= \frac{10 \times 4}{4} \\ &= 10 \text{ A} \end{aligned}$$

Step 5 of 7

Calculate the time constant τ .

$$\tau = \frac{L}{R}$$

Substitute 0.5 for L , and 4 for R .

$$\begin{aligned} \tau &= \frac{0.5}{4} \\ &= \frac{1}{8} \text{ sec} \end{aligned}$$

Step 6 of 7

The expression for the current through the inductor for $t > 0$ is,

$$i = i(\infty) + [i(0) - i(\infty)]e^{\frac{-t}{\tau}}$$

Substitute 10 for $i(\infty)$, 5 for $i(0)$, and $\frac{1}{8}$ for τ .

$$\begin{aligned} i &= 10 + [5 - 10]e^{\left(\frac{-t}{\left(\frac{1}{8}\right)}\right)} \\ &= 10 - 5e^{-8t} \text{ A} \end{aligned}$$

Therefore, the value of current i is $\boxed{(10 - 5e^{-8t})u(t) \text{ A}}$.

Step 7 of 7

The expression for the voltage across the inductor for $t > 0$ is,

$$v = L \frac{di}{dt}$$

Substitute $(10 - 5e^{-8t})$ for i , and 0.5 for L .

$$\begin{aligned} v &= 0.5 \times \frac{d}{dt}(10 - 5e^{-8t}) \\ &= 0.5(0 + 8 \times 5e^{-8t}) \\ &= 20e^{-8t} \text{ V} \end{aligned}$$

Therefore, the voltage across the inductor is $\boxed{20e^{-8t}u(t) \text{ V}}$.